

A SHARP INTERACTION-DEGREE THRESHOLD FOR SIMULATING QAOA

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ABSTRACT. We identify a sharp interaction-degree threshold for the classical simulation of QAOA with 2-local cost functions. At degree 3, classical sampling from depth-1 QAOA with small multiplicative error would collapse the polynomial hierarchy to its third level. At degree 2, exact classical sampling from depth- p QAOA on n qubits runs in time $n^{O(1)}$ whenever $p = O(\log n)$. The hard degree-3 instances have trivially optimizable cost functions, so sampling hardness does not by itself imply a quantum optimization advantage.

1. INTRODUCTION

QAOA (the Quantum Approximate Optimization Algorithm, [FGG14]) is one of the main near-term quantum algorithms. Farhi and Harrow [FH16] showed that, already at depth 1, classical sampling from the output distribution of QAOA with sufficiently small multiplicative error would collapse the polynomial hierarchy. Their result is often cited as evidence that even very shallow QAOA can exhibit behavior inaccessible to classical sampling.

We examine the structural conditions under which this hardness persists. We consider 2-local cost functions $C(x_1, \dots, x_n) = \sum_i C_i(x_1, \dots, x_n)$, where each term depends on at most two variables, and parameterize instances by their interaction degree, the maximum number of other variables coupled to any single variable. We show that depth-1 QAOA remains classically hard to sample from when the interaction degree is at most 3. Moreover, the hard instances can be chosen so that the cost function is trivial to optimize classically, so sampling hardness alone does not yield a quantum advantage for optimization.

We then show that at interaction degree 2, every marginal output probability of a depth- p QAOA circuit can be computed classically in time polynomial in n and exponential only in p . In particular, exact sampling is polynomial-time whenever $p = O(\log n)$. The simulation uses the Markov–Shi algorithm for quantum circuits of small cut width [MS08].

A comparable dichotomy holds classically for constraint satisfaction. SAT restricted to instances in which every variable occurs in at most 3 clauses remains NP-complete, but becomes polynomial-time solvable when every variable occurs in at most 2 clauses [Tov84]. The transition from 2-SAT to 3-SAT [Kar72, GJ79] is the most familiar instance of the same pattern. In the classical case the parameter separates P from NP-completeness, while for QAOA it separates classical simulability from sampling hardness.

All simulation bounds in this paper count arithmetic operations on matrix entries. Since every gate entry in our constructions lies in $\mathbf{Q}(e^{i\pi/4})$, every intermediate value in the simulation belongs to the same field and is represented by four rational coefficients whose bit lengths grow at most linearly in the circuit size. Consequently, the same asymptotic bounds hold in the bit model.

2. PRELIMINARIES

QAOA takes as input a cost function $C(x_1, \dots, x_n)$ on n binary variables together with a decomposition $C = \sum_i C_i$ into terms each depending on at most two variables. As a diagonal operator on n qubits, C acts by $C|z\rangle = C(z)|z\rangle$. The algorithm constructs the depth- p state

$$|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = e^{-i\beta_p B} e^{-i\gamma_p C} \dots e^{-i\beta_1 B} e^{-i\gamma_1 C} |+\rangle^{\otimes n},$$

where $B = \sum_{j=1}^n X_j$ is the mixing operator and $\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbf{R}^p$ are variational parameters. The state is measured in the computational basis. Since each C_i acts on at most two qubits, $e^{-i\gamma_k C}$ factors into commuting one- and two-qubit diagonal gates, and $e^{-i\beta_k B}$ into single-qubit rotations. Each 2-local term coupling variables j and k produces a two-qubit diagonal gate on those qubits, and the set of all such couplings forms a graph.

Definition 1. Given a cost function expressed as a sum of 2-local terms $C = \sum_i C_i$, the *interaction graph* of this decomposition is the simple graph on vertex set $[n]$ with an edge $\{j, k\}$ whenever some term C_i depends nontrivially on both variables j and k (that is, flipping either bit can change the value of C_i). The *interaction degree* is the maximum vertex degree of this graph. A QAOA instance specifies the decomposition, and we write *degree* for interaction degree when the meaning is clear.

In a post-selected circuit the output distribution is conditioned on a designated register yielding $|0 \dots 0\rangle$. Since there is no bound on how small this conditioning probability may be, post-selection is not physically realizable but is a theoretical tool for relating complexity classes. The output probability of such a circuit is a ratio of two event probabilities, and the relevant error notion must preserve such ratios.

Definition 2. A classical algorithm samples with *multiplicative error* $c \geq 1$ from a distribution $\mathcal{D}$ on $\{0, 1\}^n$ if it produces samples from a distribution $\mathcal{D}'$ satisfying $c^{-1}\mathcal{D}(x) \leq \mathcal{D}'(x) \leq c\mathcal{D}(x)$ for every x .

By summing the pointwise bounds, multiplicative error extends to event probabilities. A multiplicative-error sampler therefore approximates both the numerator and denominator of a post-selected acceptance probability, which is the mechanism behind the hardness proofs below.

We use two standard facts about post-selection,

$$\text{PostBQP} = \text{PP} \quad \text{and} \quad \text{PostBPP} \subseteq \Delta_3.$$

The first is due to Aaronson [Aar05], and the second to Han, Hemaspaandra, and Thierauf [HHT97]. Together with Toda's theorem, they imply that a multiplicative-error classical sampler for any circuit family whose post-selected version equals PostBQP would collapse PH to Δ_3 .

We now fix the post-selected QAOA class used in the degree-restricted statement.

Definition 3. A language L is in $\text{PostQAOA}_p^{\deg \leq d}$ if there exist a constant $0 < \epsilon < \frac{1}{2}$ and a uniform family of depth- p QAOA circuits of interaction degree at most d , each equipped with a designated output qubit and a post-selection register, such that for every input w :

- (i) the probability of the post-selection register returning $|0 \dots 0\rangle$ is nonzero;
- (ii) if $w \in L$, the output qubit is 1 with probability at least $1 - \epsilon$, conditioned on post-selection;
- (iii) if $w \notin L$, the output qubit is 0 with probability at least $1 - \epsilon$, conditioned on post-selection.

We write $\text{PostQAOA}_p = \bigcup_d \text{PostQAOA}_p^{\deg \leq d}$ for the class with no degree restriction.

3. HARDNESS AT INTERACTION DEGREE THREE

Two prior results establish sampling hardness for restricted circuit families using similar post-selection arguments. Bremner, Jozsa, and Shepherd [BJS11] proved $\text{PostIQP} = \text{PP}$, where IQP circuits have the form $H^{\otimes n} D H^{\otimes n} |0^n\rangle$ with D diagonal in the computational basis. Farhi and Harrow [FH16] extended this to QAOA by proving $\text{PostQAOA}_1 = \text{PostBQP}$. Both constructions reduce an arbitrary PostBQP circuit to the restricted family by replacing each intermediate Hadamard with a post-selected diagonal coupling to a fresh auxiliary qubit. The two constructions differ only in the single-qubit gate F that completes each step, and hence in the diagonal coupling it requires.

We construct the diagonal coupling for general F and add a preprocessing step that reduces the interaction degree from unbounded to 3. That is, we prove $\text{PostQAOA}_1^{\deg \leq 3} = \text{PostBQP}$ and obtain $\text{PostIQP}^{\deg \leq 3} = \text{PP}$ (where $\text{PostIQP}^{\deg \leq d}$ denotes the subclass of PostIQP in which D has interaction degree at most d) as a corollary.

Our general construction will transform the input circuit so that intermediate Hadamard gates are replaced by post-selected diagonal couplings, and all remaining pre-mixer gates commute and can be collected into a single QAOA phase layer. In the universal gate set $\mathcal{G} = \{H, T^\dagger, CZ\}$ [Aha03], every gate except the Hadamard is diagonal. Diagonal operators on n qubits form a commutative subalgebra, so any product of such gates remains diagonal and merges into one phase layer. The Hadamard lies outside this subalgebra, so each occurrence must be replaced by a diagonal coupling to an auxiliary qubit. To replace a Hadamard on qubit j , we prepare an auxiliary a in $|+\rangle$ and apply a diagonal two-qubit gate W to (a, j) , then apply F to j and post-select j on $|0\rangle$. Consequently, qubit a carries the logical wire, and we want to choose W so that the map from j to a acts as a Hadamard. Figure 1b illustrates the coupling substitution. For brevity, let us write $w_{ab} = \langle a, b | W | a, b \rangle$ and $r_b = \langle 0 | F | b \rangle$. The map from j to a is then $V: \mathbb{C}_j^2 \rightarrow \mathbb{C}_a^2$ defined by

$$V |b\rangle_j = \langle 0 |_j F_j W_{a,j} (|+\rangle_a \otimes |b\rangle_j)$$

and has entries $V_{ab} = w_{ab} r_b / \sqrt{2}$. The condition $V = \lambda H$ determines W entrywise.

Proposition 4. *Suppose $r_0 r_1 \neq 0$. Then $V = \lambda H$ if and only if*

$$w_{ab} = \frac{\lambda (-1)^{ab}}{r_b}. \quad (1)$$

A unitary diagonal W with this property exists if and only if $|r_0| = |r_1|$, and is then unique up to global phase.

Proof. If w_{ab} is given by (1), then $V_{ab} = \lambda (-1)^{ab} / \sqrt{2}$, so $V = \lambda H$. Conversely, $V = \lambda H$ forces $w_{ab} r_b = \lambda (-1)^{ab}$, which is (1). For the unitarity claim, $|w_{ab}| = 1$ for all a, b holds if and only if $|\lambda| = |r_0| = |r_1|$. Only the phase of λ remains free, and it acts as a global phase on W . $\square$

Remark 5. With $\lambda = 1/\sqrt{2}$, the Bremner–Jozsa–Shepherd substitution $W = CZ$ corresponds to $F = H$, and the Farhi–Harrow substitution $W = \text{diag}(1, i, 1, -i)$ to $F = \tilde{H} = e^{-i\pi X/4}$.

Theorem 6. *For every language $L \in \text{PostBQP}$, there exists a uniform family of post-selected depth-1 QAOA circuits of interaction degree at most 3 that decides L with bounded error. Equivalently,*

$$\text{PostQAOA}_1^{\deg \leq 3} = \text{PostBQP}.$$

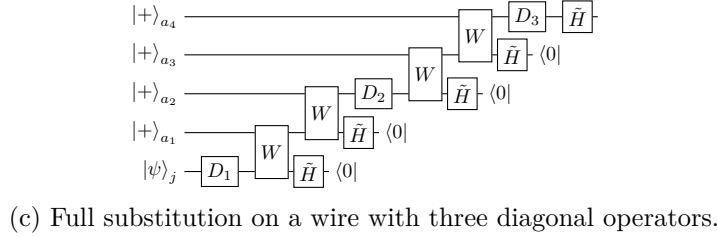
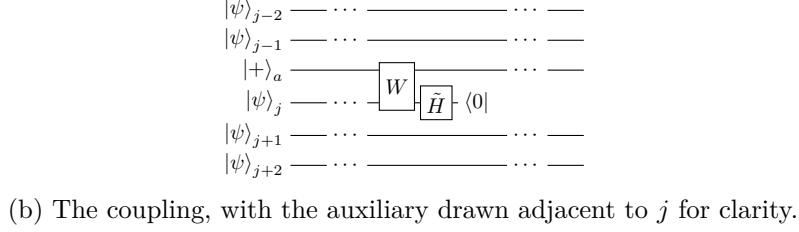
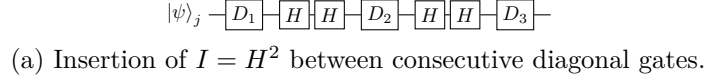


FIGURE 1. Degree reduction for three diagonal operators on a single wire.

Moreover, the reduction preserves the post-selected output distribution exactly.

Proof. The inclusion $\text{PostQAOA}_1^{\deg \leq 3} \subseteq \text{PostBQP}$ is immediate. For the reverse, we will show that decomposing over $\mathcal{G}$ and applying our coupling to each Hadamard converts an arbitrary PostBQP circuit into depth-1 QAOA, and then bound the interaction degree. For $F = e^{-i\beta X}$, Proposition 4 forces $\beta = \pi/4$ and gives $W = \text{diag}(1, i, 1, -i)$. One verifies that for all states $|\phi_0\rangle$ and $|\phi_1\rangle$,

$$\langle 0|_j \tilde{H}_j W_{a,j} (|+\rangle_a \otimes (|0\rangle_j |\phi_0\rangle + |1\rangle_j |\phi_1\rangle)) = \frac{1}{\sqrt{2}} H_a (|0\rangle_a |\phi_0\rangle + |1\rangle_a |\phi_1\rangle).$$

This identity replaces each intermediate Hadamard by a diagonal coupling to an auxiliary, but applying it directly leaves the degree unbounded, since consecutive diagonal gates on one wire all contribute edges to the same qubit. We preprocess the circuit to enforce the invariant that consecutive diagonal gates on each wire are separated by a Hadamard.

If a wire begins with a Hadamard, absorb it into the preparation $|+\rangle = H|0\rangle$. If it begins with a diagonal gate, prepend H^2 and absorb the first H into the preparation. We separate consecutive diagonal gates by inserting $I = H^2$ (Figure 1a). At the end of each wire, insert $I = \tilde{H}\tilde{H}^\dagger$, absorb $\tilde{H}$ into the mixing layer, and decompose $\tilde{H}^\dagger = H e^{i\pi Z/4} H$, producing two Hadamards separated by a 1-local diagonal. After preprocessing, every wire begins in $|+\rangle$, ends with a single $\tilde{H}$, and has at most one diagonal gate between consecutive Hadamards.

With the invariant in place, we replace each intermediate Hadamard by the coupling (Figure 1b). After all substitutions, every qubit starts in $|+\rangle$ and receives exactly one $\tilde{H}$ from the mixing layer. All remaining gates before the mixer are diagonal, with phases that are eighth roots of unity. Setting $\gamma = \pi/4$ therefore collects them into a single integer-valued cost function C . The compiled circuit is

$$\tilde{H}^{\otimes n} e^{-i\frac{\pi}{4}C} |+\rangle^{\otimes n}$$

with post-selection on some qubits, which is depth-1 QAOA with $\beta = \gamma = \pi/4$.

No gate acts on a qubit after its projection onto $|0\rangle$, so the post-selections commute with all subsequent operations and form a single register, and each coupling contributes a factor of $1/\sqrt{2}$ that cancels in the post-selected distribution. The output distribution equals that of the original circuit, and the compilation is polynomial-time.

It remains to bound the degree. Along each wire the active qubits form a chain of couplings (Figure 1c). Each coupling moves the wire to a new qubit, so each qubit participates in at most two coupling edges. Preprocessing ensured at most one diagonal gate per segment, so each qubit carries at most one original 2-local gate, contributing at most one additional edge. The maximum degree is $2 + 1 = 3$. $\square$

Corollary 7. $\text{PostIQP}^{\deg \leq 3} = \text{PP}$.

Proof. Repeat the proof of Theorem 6 with the IQP specialization $F = H$, $W = CZ$, and combine with $\text{PostBQP} = \text{PP}$. $\square$

Corollary 8. *If depth-1 QAOA circuits of interaction degree at most 3 could be sampled classically with multiplicative error $1 \leq c < \sqrt{2}$, then $\text{PH} \subseteq \Delta_3$.*

Proof. By Theorem 6, $\text{PostQAOA}_1^{\deg \leq 3} = \text{PP}$. Given $L \in \text{PP}$ and a constant $\epsilon > 0$, amplification within PostBQP [Aar05, §3] followed by Theorem 6 produces a degree-3 depth-1 QAOA circuit whose post-selected acceptance probability q satisfies $q \geq 1 - \epsilon$ on yes-instances and $q \leq \epsilon$ on no-instances. A multiplicative-error c sampler approximates every event probability within factor c , so it estimates q to within c^2 . On a yes-instance the estimate is at least $(1 - \epsilon)/c^2 > \frac{1}{2}$, and on a no-instance at most $c^2\epsilon < \frac{1}{2}$, provided $c < \sqrt{2}$ and ϵ is small enough. Thus $\text{PP} \subseteq \text{PostBPP}$ and

$$\text{PH} \subseteq \text{P}^{\text{PP}} \subseteq \text{P}^{\text{PostBPP}} \subseteq \text{P}^{\Delta_3} = \Delta_3$$

by Toda's theorem and $\text{PostBPP} \subseteq \Delta_3$. $\square$

Remark 9. Additive error would not suffice in Corollary 8. The post-selection event may have exponentially small probability, so an additive-error guarantee on the joint distribution gives no useful bound on the conditional. Extending the hardness to additive error requires an average-case conjecture, as illustrated by the IQP case [BMS16].

The hard instances of Theorem 6 have a trivially optimizable cost function.

Proposition 10. *The cost function C constructed in Theorem 6 can be made monotone and maximized at $x = 1^n$ without altering the QAOA circuit.*

Proof. Since $e^{-i(\pi/4)C}$ depends only on C modulo 8, individual diagonal entries may be shifted by multiples of 8. Write $W = e^{-i(\pi/4)C_W}$ with $C_W = 6|01\rangle\langle 01| + 2|11\rangle\langle 11|$. The $T^\dagger$ and CZ contributions $|1\rangle\langle 1|$ and $4|11\rangle\langle 11|$ already have nonnegative coefficients. Replacing the C_W diagonal $(0, 6, 0, 2)$ by the congruent $(0, 6, 0, 10)$ and the endpoint diagonal $(7, 1)$ by $(7, 9)$ makes every coefficient nonnegative, so C is monotone and maximized at $x = 1^n$. $\square$

4. TRACTABILITY AT INTERACTION DEGREE TWO

We now show that the degree-3 bound is tight. A graph of maximum degree 2 decomposes into paths, cycles, and isolated vertices, and the QAOA circuit on each component has small cut width.

Definition 11. A gate on qubits S crosses cut $i \in \{1, \dots, n-1\}$ if $\min S \leq i < \max S$. Single-qubit gates cross no cut. The *cut width* of a circuit on qubits $[n]$ is the maximum number of gates crossing any single cut.

Markov and Shi simulate a circuit by contracting its tensor network along a linear qubit ordering, at cost exponential in the number of gates crossing any single cut.

Proposition 12 (Markov–Shi [MS08, Proposition 5.1]). *A quantum circuit of size T and cut width r , in which each gate has an equal number of input and output qubits, can be simulated deterministically in time $T^{O(1)} \exp[O(r)]$.*

By simulation we mean computing $\Pr[Z_S = z_S]$ for a specified subset $S \subseteq [n]$ and outcome z_S . Unmeasured qubits are traced out during contraction at no additional asymptotic cost.

Theorem 13. *Let Q be a depth- p QAOA circuit on n qubits whose interaction graph has maximum degree 2. For any $S \subseteq [n]$ and $z_S \in \{0, 1\}^{|S|}$, the marginal probability $\Pr[Z_S = z_S]$ can be computed deterministically in time $(np)^{O(1)} \exp(O(p))$.*

Proof. Phase gates couple only qubits within the same component and mixer gates act on single qubits, so the state factors as $|\psi\rangle = \bigotimes_c |\psi_c\rangle$. It therefore suffices to simulate each component independently.

A graph of maximum degree 2 decomposes into isolated vertices, paths, and cycles. An isolated vertex sees only single-qubit gates, so its output distribution is computable in $O(p)$ operations. For a linear ordering of the vertices, let $\delta(i)$ denote the number of interaction-graph edges crossing cut i . Each such edge contributes one gate per phase layer, so the circuit cut width is $p \cdot \max_i \delta(i)$. For a path in the natural ordering, each edge $\{i, i+1\}$ crosses only cut i , so $\max_i \delta(i) = 1$ and the cut width is p . For a cycle, the wrap-around edge $\{1, k\}$ crosses every cut, so $\max_i \delta(i) = 2$ and the cut width is $2p$.

In every case the cut width is $O(p)$ and the circuit has size $O(kp)$. Proposition 12 gives runtime $(kp)^{O(1)} \exp(O(p))$ per component, and summing over components whose sizes total n gives $(np)^{O(1)} \exp(O(p))$. $\square$

Corollary 14. *The output distribution of a depth- p QAOA circuit on n qubits with interaction degree at most 2 can be sampled exactly by a classical randomized algorithm in time $(np)^{O(1)} \exp(O(p))$. When $p = O(\log n)$, this equals $n^{O(1)}$.*

Proof. Sample $z_1, \dots, z_n$ sequentially by the chain rule. At step t , two calls to Theorem 13 give $\Pr[Z_1 = z_1, \dots, Z_{t-1} = z_{t-1}, Z_t = b]$ for $b \in \{0, 1\}$, and we draw z_t from the conditional distribution. After n steps, $z = z_1 \cdots z_n$ is drawn from the exact output distribution. The $2n$ calls cost $(np)^{O(1)} \exp(O(p))$ total, which equals $n^{O(1)}$ when $p = O(\log n)$. $\square$

5. DISCUSSION

The simulation threshold for 2-local QAOA lies between interaction degrees 2 and 3. Since exact simulation implies simulation with any multiplicative error, degree-2 QAOA is classically easy under both error models when $p = O(\log n)$.

The degree-2 simulation runs in time exponential in p , and whether a different method can improve this is open. Corollary 8 assumes multiplicative error, and extending the hardness to additive error would require an anticoncentration bound and an average-case hardness conjecture for random 2-local cost functions on degree-3 graphs. The analogous results for IQP [BMS16] do not transfer because the anticoncentration proof uses completeness of the interaction graph and the average-case families have unbounded degree. The hard instances have trivially optimizable cost functions (Proposition 10), and whether there exist families for which both optimization and sampling are hard remains open.

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